

What You Rehearse Is What You Keep: Rehearsal Format Decides Whether Known Facts Stay Composable

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Abstract

Fine-tuning a language model on new facts can leave the facts it already knew intact as answers while removing its ability to use them together. Rehearsal is the standard protection, and the usual unit of rehearsal is the individual fact. We ask what rehearsal must supervise to preserve composition.

On two-hop routes that two 3B-parameter models (Qwen2.5-3B-Instruct, Falcon3-3B-Instruct) compose robustly under twelve phrasings before any update, an update with no rehearsal costs 18 and 34 percentage points of compositional access on phrasings the model never saw. Rehearsing the composition in a single phrasing restores it to 0.99. Rehearsing the two constituent facts as answers, with about twice the supervised tokens, leaves it near the no-rehearsal level (composition — facts: +0.166 [+0.032, +0.301] on Qwen, +0.204 [+0.097, +0.311] on Falcon, six seeds each). Fact rehearsal produces a specific failure — the model answers the composite question with the intermediate entity — on 8–13% of held-out phrasings against about 1% without rehearsal, and it collapses composition (held-out access below 0.7) in 3 of 18 adapters at each checkpoint, with loss curves and memorisation epochs like those of the adapters that did not collapse. On the failing phrasings the second inferential step is usable: supplying the intermediate entity’s representation restores the answer on 86% (Qwen) and 61% (Falcon) of them.

A controlled transfer experiment shows the effect is one of rehearsal format and that the protection generalises. Rehearsing 340 routes protects 320 never-rehearsed routes that share no subject or intermediate entity with them (+0.129 [+0.036, +0.222]); on a second family, +0.269 [+0.078, +0.460]. Two formats preserve composition: the composite question with its answer, and the same two facts written as context with only the final answer supervised (0.985 and 0.970 on rehearsed routes). Formats that supervise the facts themselves, as isolated answers or as a continuation of the first-hop prompt, do not (0.648 and 0.831). Declared ablations separate what the format changes. Removing the loss on the intermediate entity where it answers the first hop eliminates the bridge-as-answer failure in every seed of both families without recovering composition; keeping the same text with loss on the final answer alone does not recover it either. What separates the formats is the chat boundary: the same facts, with the same supervised span, preserve composition better as the user’s input than as the model’s own response, in every declared seed of both families; a seed extension decided after that result gives six-seed intervals excluding zero in each (+0.099 [+0.045, +0.152] on Qwen, +0.091 [+0.019, +0.163] on Falcon), though the added seeds alone do not. **Composition survives rehearsal when the example has the shape of use — the compositional information as the user’s input, the final answer as the model’s first output; supervising the intermediate entity adds a separate failure, answering with it.**

1 · Introduction

Continual updating of a deployed model has a familiar cost: what it learns next can degrade what it knew before. Rehearsal — mixing previously learned material into the update — is the oldest and most reliable mitigation, and its unit is almost always the individual example or fact. That choice is rarely examined. If a model’s knowledge of A and of B is preserved, its ability to use A and B together is assumed to follow.

The question. What must rehearsal supervise to preserve a model’s ability to compose facts it already knows?

We answer it in the cleanest setting we could construct. The task is factual two-hop composition: *the country where the author of novel X was born* requires resolving an intermediate entity (the author) and then a fact about it (birth country). We take routes the base model already composes reliably across six phrasings and two personas, inject unrelated new facts about the intermediate entities by LoRA fine-tuning, and compare rehearsal conditions at matched content-token budgets and matched acquisition of the injected facts. The outcome is compositional access on phrasings and a persona the model never saw during the update.

What the failure looks like. The same route and phrasing, answered by the model after an update with fact rehearsal and after one with composition rehearsal (acquisition-matched checkpoint, seed 0, verbatim model outputs; neither phrasing was rehearsed):

model	held-out phrasing	fact rehearsal	composition rehearsal
Qwen2.5-3B	Name the capital of the country led by president Daniel Ortega:	Nicaragua	Managua
Qwen2.5-3B	The name of the country the city where Kevin Love was born is in is	Santa Monica	United States of America
Falcon3-3B	Name the country where the singer of ‘Party in the U.S.A.’ was born:	Miley Cyrus	United States of America
Falcon3-3B	‘The city where the spouse of Cho Yu-min was born’ refers to	Park So-yeon	Seoul

Under fact rehearsal the model answers with the intermediate entity, or stops one step short of the answer.

The answer, in four parts. (§4) An update with no rehearsal costs 18–34 points of held-out compositional access; rehearsing the composition restores it; rehearsing the constituent facts as answers does not, despite roughly double the supervised tokens. (§5) Fact rehearsal induces a specific failure — the intermediate entity emitted as the final answer — that is absent without rehearsal, and it collapses composition in a minority of training seeds. (§6) On the failing phrasings the second step is usable when the intermediate entity is supplied: the failures are a missing intermediate entity, not a broken second step. (§7) The protection transfers to routes never rehearsed, in both families. Three declared ablations then take the format apart: supervising the intermediate entity as the first-hop answer produces the bridge-as-answer failure; the incremental effect of loss on the restated facts is not resolved; and writing the facts as the model’s own response instead of the user’s input lowers composition, even when only the final answer is supervised.

The practical rule is short. **Rehearse in the shape of use.** Put the compositional information in the user’s turn and make the final answer the model’s first output: the composite question with its answer, or the facts as context followed by the answer. The same facts written as the model’s own response do not protect composition, even with no loss on them, and supervising the intermediate entity as an answer additionally teaches the model to answer with it.

Contributions

1. A controlled comparison of rehearsal units — nothing, constituent facts, the composition — at matched content tokens and matched acquisition, on routes the model composes before the update, across two model families and six seeds each, with the exposure asymmetry reported in full (§4).
2. A reproducible failure signature of fact rehearsal, the intermediate entity emitted as the answer, shown to be induced by the rehearsal rather than by the update, with an operational definition of the collapse it produces and its frequency (§5).
3. A transfer experiment on 320 (Qwen) and 541 (Falcon) routes disjoint in subject and intermediate entity from every rehearsed route, with a comparison of rehearsal formats at identical facts (§7), supported by a failure-matched causal test with thresholds declared in advance (§6).
4. Three declared ablations that hold text, rows and schedule fixed and vary one thing at a time — loss on the intermediate entity, loss on the restated facts, and the position of the chat boundary — showing that the position of the chat boundary separates the formats that keep composition from those that lose it, in every declared seed and with six-seed support from a labelled post hoc extension (§7).

What this paper does not claim

The routes in the comparison set are composed *latently*: the base model names the intermediate entity on request for 30% of Qwen’s and 2% of Falcon’s comparison routes. Fact rehearsal therefore mostly *teaches* standalone

access rather than preserving it, and the claim is that fact rehearsal does not ensure compositional retention, not that preserving already accessible facts is insufficient (§8). The ablations (§7) identify an effect of chat format among formats with identical text and loss; they do not say why the model’s own turn behaves differently from the user’s, and they do not resolve whether loss on the restated facts contributes. The six-seed intervals for the chat-boundary effect come from a seed extension decided after the three-seed result; the added seeds alone do not exclude zero, so the combined result supports the effect rather than establishing it independently. §6 shows that failing computations are recoverable under intervention; it does not show that every failing computation has an undamaged second hop. We make no claim about the timing of the model’s own resolution of the intermediate entity. Two 3B models with LoRA adapters, one dataset, two-hop routes.

2 · Related work

Rehearsal and replay. Interleaving old material into new training is the classic remedy for catastrophic forgetting [Robins 1995; Rolnick et al. 2019]; the unit rehearsed is the training example. We hold the method fixed and vary what is supervised.

Forgetting as lost access rather than lost knowledge. Zheng et al. [2025] show that performance drops in continual learning of language models often reflect lost task alignment rather than lost knowledge. Our no-rehearsal baseline is a case: the facts remain recallable and the composition does not. Our contribution is which rehearsal format restores it and why the obvious one does not.

Atomic knowledge without composition. Yu et al. [2026] show that post-training recipes with indistinguishable atomic knowledge differ by over 40 points in multi-hop composition. We observe the same dissociation with the recipe fixed and only the rehearsed material varied, and identify the format property that tracks it.

Supervising intermediates. Two lines bear directly on our claim. In algorithmic reasoning with depth-recurrent transformers, supervising every intermediate step is harmful relative to a silent objective [Chen 2026]; internalised chain-of-thought removes intermediate supervision progressively [Deng et al. 2024]. Closest to our setting, Karmim et al. [2026] pretrain small models on synthetic biographies and find that training text stating a two-hop fact without its bridge supports composition far better than text naming the bridge, that combining the two is best, and that composition never reaches individuals absent from compositional pretraining. Our finding is the fine-tuning-stage counterpart, for knowledge a pretrained model already composes: it adds transfer, a causal test, and ablations that separate supervising the bridge, supervising the restated facts, and the side of the chat boundary on which the facts sit. Conversely, Lin et al. [2025] add a zero-hop identity supervision on the bridge entity (“identity bridge”) to *enable* out-of-distribution two-hop generalisation, and report that correct two-hop predictions coincide with a direct subject-to-answer association. That raises the question whether protection in our setting is genuine composition or a shortcut; §7 tests it causally on the never-rehearsed routes.

Latent multi-hop reasoning. Yang et al. [2024] and Biran et al. [2024] establish that two-hop questions are answered by resolving the intermediate entity at the subject position in early-to-middle layers, and that patching that representation can repair failures. The instrument in §6 is theirs; we use it to test availability, not timing. Balesni et al. [2024] find that models fine-tuned on two synthetic facts fail to compose them latently unless the facts co-occur in training or the prompt, but can compose one synthetic and one natural fact; our routes are pretrained knowledge on both hops, and our manipulation is supervision format at fixed content. O’Neill [2026] finds that invented facts written into the weights compose with world knowledge less reliably than the same facts supplied in the prompt at test time. Our placement result concerns training instead: rehearsed with identical text and loss, facts in the user’s turn preserve composition better than facts in the model’s own turn.

Circuit preservation and knowledge editing. Comparisons of training methods by how much of the base circuitry survives [Laitinen-Fredriksson Lundstrom-Imanov 2026; Rojas Nunez et al. 2026] and editing methods that make a new fact reach its multi-hop uses [Zhao et al. 2026; Yang et al. 2025; He et al. 2026] are adjacent; we compare rehearsal targets and ask whether an unrelated update leaves existing composition intact.

3 · Setup

3.1 Routes the model already composes

Routes are TwoHopFact chains [Yang et al. 2024]: subject $\rightarrow$ intermediate entity (“bridge”) $\rightarrow$ answer. A route enters the study only if the base model answers the composite question under all six phrasings (canonical, named, question, imperative, “known as”, “refers to”) under a domain persona and under a neutral persona, and is not answered when its first entity is replaced by a different entity in the same question frame. That control removes routes answered from the frame or from an answer prior; it does not rule out a memorised association from the correct subject to the answer, which §7 addresses with a causal test. The screen covers all 45,595 TwoHopFact chains; 4,475 (Qwen) and 3,176 (Falcon) pass the domain-persona criterion, 1,605 and 2,254 also pass the neutral persona, and after requiring a well-formed injectable fact about the bridge 753 and 1,639 routes enter the design, over 655 and 1,427 distinct bridges, of which 660 and 1,112 are measurable in the half used here. Knowledge of the second hop when asked directly is measured but is **not** a selection criterion: on the comparison set, the base model answers all six direct second-hop phrasings for 179 of 229 Qwen routes and 231 of 414 Falcon routes, and none of them for 21 and 117. An exploratory restriction to routes whose second hop is known on all six phrasings leaves the primary result unchanged in direction and excluding zero (Table S11; §4).

3.2 The update and the within-model split

For each route, an unrelated fact about the bridge entity, drawn from Wikidata and verified absent from the base model, is injected by LoRA fine-tuning [Hu et al. 2022] (rank 16, α 32, dropout 0.05, all attention and MLP projections, AdamW lr 2e-4, weight decay 0.01, micro-batch 1 with gradient accumulation over 8 rows, 12 epochs, answer-only loss, chat template with the domain persona). The incomplete last accumulation group of each epoch is discarded. Routes are split into two counterbalanced halves so that one adapter injects facts about half the bridges and every route is measured in the same adapter as its own control group; the analyses here use half X. The split makes injected and non-injected routes comparable within one adapter, where the update’s general effect is shared. The rehearsal contrasts in this paper are different: they compare different adapters, paired only by training seed, and their general effects differ (§7), which is why every one of them is reported against a no-rehearsal adapter as well.

3.3 Rehearsal conditions

Rows are appended to the injection set; the update is otherwise identical. **none**: the injected facts only. **atomic**: hop-1 prompt $\rightarrow$ bridge and hop-2 prompt $\rightarrow$ answer. **route**: the composite question in its canonical phrasing $\rightarrow$ answer. Rehearsal sets are matched on content tokens by subsampling rows, not routes, to a token budget (on Qwen, the budget of a control set of unrelated Wikidata facts; on Falcon, the atomic set’s own). On Qwen this leaves 21 of the 229 comparison routes with only one of their two facts rehearsed under the atomic condition; on Falcon all 414 have both. Restricting to the 208 Qwen routes with both facts gives +0.171 [+0.036, +0.307] at six seeds against +0.166 on all 229 (exploratory; Table S11). Because atomic rows are shorter, the atomic condition receives $1.7\text{--}2.1\times$ the supervised tokens and 17–21% more optimizer steps per epoch (Table S4). The comparison set is the routes rehearsed in both conditions: 229 on Qwen, 414 on Falcon. §7 adds two formats with identical facts.

3.4 Outcome, checkpoints and estimator

Held-out compositional access is the fraction correct over five non-canonical phrasings under the domain persona and six phrasings under the neutral persona; the canonical phrasing under the domain persona is the rehearsed string and is excluded. Scoring is answer-or-alias containment after lower-casing and stripping punctuation; word-boundary and exact-match variants applied to the five domain-persona predictions change the primary contrast by at most 0.004 (Table S1). **Bridge emission** is the fraction of the five domain-persona predictions whose wrong answer contains the bridge entity.

The **primary checkpoint** is the first epoch at which a fixed 40-item probe of the injected facts reaches at least 99% strict recall (“acquisition-matched”); full-set recall over all injected routes at that checkpoint is reported per condition (0.97–0.98 in both families for the primary comparison, minimum 0.93; Table 1 and Table S10). The **secondary checkpoint** is the fixed 12-epoch schedule, reported alongside. The unit of replication is the training seed; contrasts are paired over seeds with $t(k-1)$ intervals.

3.5 Declared before the run

Each experiment’s design, predictions and thresholds were written and dated before its runs; all are listed with their outcomes in PREDICTIONS.md. Nothing was deposited with a third-party registry. Three seed extensions, from three seeds to six, were decided after the three-seed intervals were seen and are reported as such (§4, §7).

4 · The unit of rehearsal determines what is preserved

Table 1. Held-out compositional access on the comparison set, acquisition-matched checkpoint. Route and atomic: six seeds; none: seeds 0–2, and contrasts against it pair those three seeds (results/tables/T1_preservation.md, T2_no_rehearsal.md).

	none	atomic	route	route — atomic (6 seeds)	route — none (3 seeds)	atomic — none (3 seeds)
Qwen2.5-3B	0.817	0.825	0.991	+0.166 [+0.032, +0.301]	+0.173 [+0.080, +0.267]	+0.055 [−0.108, +0.217]
Falcon3-3B	0.659	0.786	0.990	+0.204 [+0.097, +0.311]	+0.334 [+0.207, +0.461]	+0.101 [−0.301, +0.504]

Per-seed route — atomic: Qwen +0.157 / +0.107 / +0.092 / +0.062 / +0.164 / +0.416; Falcon +0.176 / +0.403 / +0.118 / +0.147 / +0.184 / +0.195. The declared three-seed results were +0.119 [+0.035, +0.203] (Qwen) and +0.232 [−0.143, +0.607] (Falcon; every seed positive, interval not excluding zero); both extensions to six seeds were decided after those intervals were seen, and Falcon seeds 3–5 alone give +0.175 [+0.113, +0.238]. At the fixed 12-epoch schedule: +0.126 [−0.006, +0.258] and +0.183 [+0.098, +0.267]. Full-set injected recall at the primary checkpoint: route 0.979 / atomic 0.972 (Qwen), 0.976 / 0.978 (Falcon).

An update with no rehearsal costs 18 (Qwen) to 34 (Falcon) points of compositional access on phrasings the model never saw. Rehearsing the composition in one phrasing returns it to 0.99, and the advantage is present on every held-out phrasing and tightest on the least similar one (Jaccard 0.60 to the rehearsed string), so it is not memorisation of the string. Rehearsing the constituent facts preserves the facts — first-hop access 0.95–0.98 under atomic rehearsal against 0.02–0.31 under route rehearsal — and leaves composition near the no-rehearsal level: the atomic — none intervals span zero in both families at three seeds, which does not establish equivalence but does show that no protection was detected.

Exposure is not matched beyond content tokens: atomic rehearsal supervises $1.7\text{--}2.1\times$ as many tokens and takes more optimizer steps both per epoch and to the acquisition-matched checkpoint (Table S4). The route advantage therefore does not come from more rehearsal signal, but extra supervised updates need not help monotonically, so the asymmetry is not excluded as a source of additional drift under atomic rehearsal. Retention-perplexity changes overlap between conditions (every adapter’s value is in Table S9). On the comparison set the base model names the bridge on request for 30% of Qwen routes and 2% of Falcon routes, so atomic rehearsal largely *teaches* standalone access; on the 51 Qwen routes where the bridge is already at ceiling, the effect is the same size and, at six seeds, excludes zero (+0.138 [+0.008, +0.268]). Restricting to routes whose second hop the base model answers under all six direct phrasings (exploratory; Table S11) gives +0.148 [+0.007, +0.289] on Qwen (179 routes) and +0.175 [+0.047, +0.302] on Falcon (231 routes).

5 · Fact rehearsal induces a specific failure that sometimes collapses composition

Table 2. Bridge emission on the five held-out domain-persona phrasings, comparison set, acquisition-matched checkpoint; none three seeds, route and atomic six.

	none	route	atomic (per seed)
Qwen2.5-3B	0.012	0.004	0.129 (0.112 / 0.072 / 0.074 / 0.028 / 0.145 / 0.341)
Falcon3-3B	0.001	0.000	0.078 (0.042 / 0.191 / 0.027 / 0.036 / 0.105 / 0.069)

Under atomic rehearsal the dominant wrong answer to the composite question is the intermediate entity itself: *the country where the author of X was born* $\rightarrow$ the author. It accounts for over half of all misses on Qwen in every seed. The rate is about 1% or less with no rehearsal and under route rehearsal; the failure is induced by rehearsing the bridge as an answer, not by the update, and removing the loss on the bridge where it answers the first-hop prompt removes it (§7). (Falcon adds a smaller second mode, a city offered where a country was asked, absent on Qwen; Table S2.)

The failure sometimes collapses composition. Defining collapse as held-out access below 0.7 on the comparison set, **3 of the 18 atomic-rehearsal adapters in this paper collapse at the acquisition-matched checkpoint and 3 of 18 at the fixed schedule** (0.32–0.59 and 0.38–0.67), against 0 of 18 route adapters at either (Table S9). The remaining atomic adapters lie between 0.705 and 0.928, so the count depends on a threshold placed in a narrow gap. Collapsed adapters’ memorisation epochs lie within their siblings’ range; their retention-perplexity change lies within it for three of the six and outside it for three, above in two cases and below in one, so neither is an indicator of collapse in these data. The instability is a property of the condition, and it is why every contrast against atomic rehearsal has a wider interval than its consistent sign suggests.

6 · On the failing phrasings the second step is usable

What is missing when a composition fails under fact rehearsal: the intermediate entity, or a working second step? We tested this on the computations that actually fail, with thresholds declared before the run (below 15% rescue would support a broken second step; at or above 30%, a missing entity).

For every held-out phrasing the atomic-rehearsal adapter answers wrongly (Qwen 194 / 153 / 110, Falcon 375 / 904 / 279, seeds 0–2), the residual stream at the subject’s final token at a mid-depth layer (Qwen 12 of 36, Falcon 11 of 22) was overwritten during prefill with the bridge entity’s representation read from a prompt that names it; with a different route’s bridge; or with the same position read from the bridge-prefixed prompt as a context control. The unpatched pass reproduced the stored miss on 99.6–99.8% of prompts.

Table 3. Fraction of failing phrasings answered correctly after the patch; seed means with t(2) intervals (T5_rescue.md).

donor	Qwen2.5-3B	Falcon3-3B
own bridge	0.857 [0.733, 0.980]	0.610 [0.310, 0.911]
context control	0.037	0.103
wrong bridge $\rightarrow$ that route’s answer	0.564 (on successes: 0.58)	0.458 (on successes: 0.53)

Every seed in both families clears the upper threshold. Handed the bridge, the model answers on most failing phrasings; handed a wrong bridge, it follows it to the wrong route’s answer about as often as on succeeding phrasings. The second step is usable when the bridge is supplied at the subject position; on the remaining 14% (Qwen) and 39% (Falcon) of failures it is not, and those are not characterised further here. (The competing account — a damaged second step, suggested by screens of computations that succeed — is the one the declared thresholds were set to distinguish; both accounts, the screens, and a further experiment on whether the base model’s own state at that site is sufficient are in Supplementary S1–S3.)

7 · The protection transfers, and composition depends on where the facts sit

Two objections remain after §4–§6. Rehearsing the composite question and testing its paraphrases might be ordinary rehearsal of the tested association; and isolated question–answer rows might be an unfair way to rehearse facts. One experiment, run on both families, answers both.

Design. The 660 measurable Qwen routes were split once into 340 eligible for rehearsal (R) and 320 never rehearsed in any condition (L), with routes sharing a subject or a bridge clustered so that no entity appears on both sides. Four rehearsal sets were drawn from R at matched content tokens (within 0.3%): **route** (340 routes); **atomic** (238 routes, two rows each); **coherent** — hop-1 prompt → “*bridge. hop-2 statement answer*”, both facts supervised in one continuation (232 routes); and **bridge-as-context** — “*hop-1 prompt bridge. hop-2 prompt*” → answer, the same two facts with the bridge only ever in the prompt (232 routes). Coherent and bridge-as-context contain identical facts; they differ in where the chat boundary falls and therefore in whether the bridge and the second-hop statement are inside the supervised output. Three seeds; predictions declared in advance. The Falcon replication uses the same builder (571 rehearsed-eligible, 541 never-rehearsed) with all four formats. **Loss-mask ablation.** A fifth condition, declared separately after the rest were in (PREDICTIONS.md, G16), uses the coherent rows exactly — same prompt, same completion, same chat boundary, same rows and schedule — but gives no loss to the leading bridge tokens, the ones that answer the first-hop prompt. Its input tokens are identical to the coherent set’s in every row; the masked span averages about four tokens. The bridge is then restated at the start of the second-hop statement (“... . Gustave Flaubert was born in ...”), and that copied mention keeps its loss in every row. The ablation therefore separates supervising the bridge *as the retrieved answer to the first hop* from everything else in the coherent format; it does not remove every supervised occurrence of the bridge. A sixth condition, **answer-only**, declared after the fifth was in, keeps the coherent text and gives loss to the final answer alone. It then differs from bridge-as-context only in where the chat boundary falls — the facts are the model’s own response instead of the user’s input — with the same supervised span, the final answer; it differs from the masked condition only in the loss on the restated facts. Its input tokens equal the coherent set’s in every row, and its supervised span decodes to the final answer in every row.

Table 4. Held-out compositional access and bridge emission, acquisition-matched, seeds 0–2 (T6_transfer.md). The coherent, bridge-masked and answer-only formats share their text and differ only in which tokens receive loss.

Qwen2.5-3B	none	atomic	coherent	bridge masked	answer only	bridge-as-context	route
L: 320 never-rehearsed routes	0.819	0.743	0.858	0.879	0.890	0.947	0.948
R∩: 232 routes rehearsed in all	0.762	0.648	0.831	0.884	0.866	0.970	0.985
bridge emission on R∩	0.011	0.287	0.049	0.001	0.016	0.005	0.003

Falcon3-3B	none	atomic	coherent	bridge masked	answer only	bridge-as-context	route
L: 541 never-rehearsed routes	0.654	0.893	0.763	0.767	0.798	0.905	0.923
R∩: 375 routes rehearsed in all	0.681	0.850	0.808	0.791	0.831	0.948	0.972
bridge emission on R∩	0.003	0.048	0.007	0.000	0.001	0.000	0.000

contrast on the never-rehearsed routes	Qwen	Falcon
route — none	+0.129 [+0.036, +0.222], 3/3	+0.269 [+0.078, +0.460], 3/3
bridge-as-context — none	+0.128 [+0.052, +0.204], 3/3	+0.251 [+0.111, +0.391], 3/3
route — bridge-as-context	+0.001 [−0.025, +0.027]	+0.018 [−0.036, +0.073]
atomic — none	−0.076 [−0.696, +0.543]	+0.238 [+0.041, +0.436], 3/3
bridge-as-context — coherent	+0.089 [−0.003, +0.181], 3/3	+0.142 [−0.087, +0.372], 3/3
route — atomic	+0.206 [−0.365, +0.776], 3/3	+0.031 [−0.091, +0.152]

Transfer. Rehearsing routes raises compositional access on routes that share no subject or bridge with them, in every seed of both families, with intervals excluding zero. Most of the benefit is general, not confined to the rehearsed association. The second family adds a nuance: on Falcon, fact rehearsal of *other* routes also transfers general protection (0.893 against 0.923 under route rehearsal), so the damage the atomic format does is local to the routes whose intermediate entity it supervises, while the general drift of the update is repaired by any on-distribution rehearsal. On Qwen, in the two atomic seeds that did not collapse, atomic rehearsal also exceeds no rehearsal on the never-rehearsed routes (0.878 and 0.873 against 0.829 and 0.787).

Format. On the rehearsed routes, the two formats that give the compositional information in the prompt and supervise only the final answer, route and bridge-as-context, preserve composition in both families (0.985 and 0.970 on

Qwen, 0.972 and 0.948 on Falcon). The formats that supervise the facts do not: atomic (0.648, 0.850) and coherent (0.831, 0.808). Bridge-as-context exceeds atomic in every seed of both families (+0.323 on Qwen; +0.098 [−0.073, +0.269] on Falcon) and exceeds coherent in every seed of both families (+0.139 [−0.005, +0.283] on Qwen; +0.140 [−0.071, +0.350] on Falcon). Atomic and coherent are not consistently ordered: coherent is above atomic in every Qwen seed and below it on Falcon (−0.041 [−0.173, +0.090]). Atomic rehearsal’s position relative to no rehearsal differs by family: below it on Qwen (0.648 against 0.762, driven by a collapsed seed) and above it on Falcon (0.850 against 0.681). The declared emission prediction — emission under bridge-as-context below half of atomic, with higher held-out access — was met in every seed of both families.

The loss-mask ablation separates the failure’s signature from the composition loss. Removing the loss on the leading bridge tokens eliminated bridge emission: 0.049 → 0.001 on Qwen and 0.007 → 0.000 on Falcon, below half in every seed of both families, as declared. It also removed what supervising the bridge had taught: standalone first-hop access fell to its no-rehearsal level (masked — coherent −0.670 on Qwen, −0.819 on Falcon). At the acquisition-matched checkpoint it did **not** recover the answer-only formats’ composition. Masked — coherent on the rehearsed routes was +0.053 [−0.093, +0.198] on Qwen (2 of 3 seeds) and −0.017 [−0.156, +0.122] on Falcon (1 of 3), against a declared criterion of improvement in every seed; bridge-as-context stayed above the masked format by +0.086 [+0.064, +0.109] and +0.157 [+0.032, +0.282], in every seed. (At the fixed schedule masking helps on Qwen, +0.079 [+0.026, +0.131] in every seed, bringing it within +0.019 of bridge-as-context, and hurts on Falcon, −0.121 [−0.288, +0.045]; Table S6.) The declared prediction — that removing this supervision recovers composition in every seed — failed, and the gap to bridge-as-context that remains excludes zero in both families. That does not show the supervision has no effect: on Qwen the masked format closes +0.053 of the +0.139 coherent-to-bridge-as-context gap, the intervals allow a larger share, and at the fixed schedule it closes most of it. The supported statement is narrower. Supervising the intermediate entity as the answer to the first-hop prompt is the source of the bridge-as-answer failure and of standalone access to the first hop, and at the acquisition-matched checkpoint it accounts for at most part of the composition loss, in a checkpoint-dependent way. What still separates the masked format from bridge-as-context is where the facts sit: in the masked format the second-hop statement, including a copied mention of the bridge, is written in the assistant turn and supervised as a continuation of the first-hop prompt; in bridge-as-context it is part of the prompt and only the final answer is supervised. This design does not separate the chat boundary from the supervised continuation, nor the copied mention of the bridge from the rest of it.

Moving the chat boundary separates the formats. The answer-only condition separates the two (Table S12). Its decision rule, declared before the run, measured the share of the gap between the masked format and bridge-as-context that answer-only loss closes: at least two thirds in both families would support the supervised continuation as the cause, at most one third the placement of the chat boundary. It closed −0.21 of the gap on Qwen and +0.26 on Falcon, and bridge-as-context stayed above answer-only in every seed of both families (+0.105 [−0.027, +0.236] and +0.117 [−0.039, +0.272]). The rule was declared on direction, and each family’s three-seed interval includes zero. A seed extension decided after that result, and labelled post hoc, ran seeds 3–5 of both conditions in both families under a rule fixed before those runs: the effect counts as established in a family if its six-seed interval excludes zero. It does in both: +0.099 [+0.045, +0.152] on Qwen (6 of 6 seeds) and +0.091 [+0.019, +0.163] on Falcon (5 of 6). Seeds 3–5 alone point the same way but do not exclude zero by themselves (+0.093 [−0.056, +0.242] and +0.066 [−0.125, +0.258]; one Falcon seed is at zero), so the combined result supports the placement account rather than establishing it independently. The incremental effect of loss on the restated facts, including the copied bridge, is not resolved (answer-only — masked −0.018 [−0.167, +0.130] on Qwen, +0.040 [−0.019, +0.100] on Falcon). What the data support is an effect of chat format: with identical text, identical rows and the final answer as the only supervised span, the same facts preserve composition better when they are the user’s input than when they are the model’s own response. Why the two turns differ is not tested. The never-rehearsed routes agree (+0.049 [−0.000, +0.099], 6 of 6; +0.089 [+0.006, +0.172], 6 of 6), and so does the fixed 12-epoch checkpoint (+0.082 [+0.014, +0.151], 5 of 6; +0.108 [+0.029, +0.186], 6 of 6), where the three-seed estimates had been weaker (Table S12).

The three ablations decompose the format effect. Supervising the intermediate entity as the answer to the first hop produces the bridge-as-answer failure and standalone access to the first hop; the incremental effect of supervising the rest of the restatement is not resolved; and placing the facts in the model’s own turn instead of the user’s lowers composition. Composition survives rehearsal when the example has the input–output shape of use: the compositional information as the user’s input, the final answer as the model’s first output.

No sign of a shortcut. On the never-rehearsed routes, transplanting a wrong bridge at the subject position changes the answer under route and bridge-as-context rehearsal at least as much as in the base model (identity +0.597 and +0.682 against +0.503; the declared band for “genuine composition” was ± 0.10 of base, and bridge-as-context lies above it, in the direction opposite to a shortcut; Table S7). This test covers Qwen, one seed and one layer. Within that scope the protected answers still depend on the bridge representation; it does not exclude a subject-to-answer association carried elsewhere in the network or in other seeds.

8 · Limitations

Two 3B models with LoRA adapters; one dataset; two-hop routes only. The comparison-set routes are composed latently (§4), so “preserving known facts” would misdescribe the atomic condition; nor was knowledge of the second hop a selection criterion (§3.1), though restricting to routes where it is fully known leaves the primary result in place (Table S11). The acquisition-matched checkpoint is defined on a 40-item probe; full-set recall at that checkpoint is 0.97–0.98 for the primary comparison but 0.89 for one Falcon transfer adapter, and the fixed-schedule results are reported beside every primary contrast for that reason. Exposure is matched on content tokens only (Table S4); the asymmetry disfavors the route condition. Three seed extensions were decided post hoc and are labelled; one route seed dipped at the fixed schedule only (0.81) and is retained. The format comparison and the three ablations run on both families; the chat-boundary effect meets its declared rule at three seeds, and its per-family six-seed intervals come from a post hoc extension. Why the model’s own turn behaves differently from the user’s is not tested. Never-rehearsed routes share relation types with rehearsed ones by design, so transfer is across entities within relation types. §6 establishes recoverability under intervention at one position and layer, not the timing or the full extent of the damage.

A prediction for larger models. The bridge-as-answer failure operates through the training objective — supervising a token as a target strengthens the prompt-to-token mapping at the output — and the ablation ties it to supervising the intermediate entity as the answer to the first hop, so we expect it at every scale. The chat-boundary effect concerns how an instruction-tuned model separates the user’s input from its own output, which larger instruction-tuned models share, so we expect its direction to hold as well. We expect the *magnitude* and the collapse frequency to shrink with scale if pathway redundancy grows with depth: the shallower family here (Falcon, 22 layers) showed more distributed damage and a lower explicit-bridge rescue than the deeper one (Qwen, 36 layers; 61% vs 86%), consistent with more places for the same interference to spread across. A larger-model replication would decide between the two readings; an effect of undiminished size at 70B would be the more surprising result, because it would mean the interference is not a capacity phenomenon.

9 · Conclusion

Rehearsal preserves what it rehearses in the shape it is used. On routes a model already composes, rehearsing the constituent facts as answers preserves the facts and not the composition, produces a specific failure in which the intermediate entity is emitted as the answer, collapses composition in a minority of seeds, and does so despite more supervised signal than rehearsing the composition. On the failing phrasings the second step is usable once the intermediate entity is supplied. Rehearsal format decides whether known facts stay composable: the composite question, and the same two facts written as context with only the final answer supervised, preserve composition and transfer the protection to routes never rehearsed, in two model families; formats that supervise the facts themselves do not. Taken apart, the format effect has two components: supervising the intermediate entity as an answer makes the model answer with it, and placing the facts in the model’s own turn instead of the user’s lowers composition even when only the final answer is supervised, an effect seen in every declared seed and supported by a post hoc six-seed extension. The rule that follows: rehearse in the shape of use, with the compositional information as the user’s input and the final answer as the model’s first output.

Supplementary material

S1 · Causal screens on successful computations. Before the failure-matched test, the transplant instrument was run on the canonical prompt of every comparison-set route each adapter still answers correctly, at three layers per family, with the bridge representation read from a prompt that names it and a wrong bridge from a distinct-answer route of the same template (Table S8). Identity of the bridge representation (own — wrong) under atomic rehearsal sits above base at every layer in both families (Qwen +0.576 / +0.668 / +0.659 against base +0.449 / +0.498 / +0.626 at layers 6 / 12 / 24; Falcon +0.497–0.547 against +0.393–0.396 at layers 7 and 11, three seeds), and route rehearsal at or below base on Falcon (route — atomic −0.176 [−0.296, −0.056] and −0.212 [−0.393, −0.031] at layers 7 and 11). The “answer follows the patch” statistic was first computed loosely (any same-template answer accepted)

and is stored in the result files both loosely and strictly (the actual donor’s answer); strict following under atomic rehearsal exceeds route rehearsal at every layer on both families.

S2 · The two accounts the failure-matched test was declared to distinguish. The screens in S1 show atomic rehearsal *raising* the answer’s dependence on the bridge representation on computations that succeed. One reading is that the bridge is resolved and the second hop is damaged; the other that the bridge is unavailable on the phrasings that fail. Because S1 only examines successes, it cannot decide. The test in §6 was declared with thresholds (rescue below 15% for the first account, at or above 30% for the second) and decided for the second in every seed of both families.

S3 · Natural-activation transfer. The same failing phrasings were patched with the base model’s own state at the subject position for the same unaided prompt, with no bridge named anywhere (Table S5). It rescues 0.252 [0.183, 0.321] (Qwen) and 0.363 [0.235, 0.490] (Falcon) of failures, against 0.117 and 0.204 for a wrong subject’s state and 0.063 and 0.278 for the base state at the last prompt token; the route adapter’s state at the subject position rescues 0.304 and 0.379. The natural state is partially sufficient and subject-specific in both families, and position-specific on Qwen only; 82–94% of the phrasings it rescues are ones the explicit bridge also rescues. Declared thresholds and their outcomes are in PREDICTIONS.md.

S4 · Exposure accounting (Table S4). Rehearsal rows per epoch: Qwen route 461 rows, 8,740 content tokens, 22,654 processed, 1,520 supervised; Qwen atomic 637 rows, 8,738 / 28,048 / 2,644; Falcon route 819 rows, 14,926 / 42,807 / 1,878; Falcon atomic 1,112 rows, 14,929 / 52,937 / 4,025. Optimizer steps per epoch count all training rows, the injected facts (390 Qwen, 831 Falcon) plus the rehearsal rows: 106 vs 128 (Qwen), 206 vs 242 (Falcon). Mean steps to the acquisition-matched checkpoint: 707 vs 875 (Qwen), 1,785 vs 2,138 (Falcon).

S5 · Scoring-rule variants (Table S1). Applied to the five domain-persona predictions, the study rule, substring, word-boundary and exact-match give +0.119 / +0.119 / +0.119 / +0.123 (Qwen) and +0.232 / +0.232 / +0.232 / +0.233 (Falcon) for the three-seed primary contrast.

S6 · Subgroups (Table S3). Never-rehearsed routes in the preservation experiment (Qwen 98, Falcon 151): route — atomic +0.080 [−0.028, +0.188] and +0.123 [+0.027, +0.219] at six seeds. Own-atoms difference-in-differences, (route — atomic on routes whose atoms only were rehearsed) — (route — atomic on never-rehearsed routes): +0.068 [+0.019, +0.116] (Qwen), +0.040 [+0.000, +0.079] (Falcon). Full-set injected recall per adapter and checkpoint is in Table S10; the lowest value in any primary comparison is 0.888 (one Falcon bridge-as-context adapter at its acquisition-matched checkpoint), and 0.987 for the same adapter at the fixed schedule.

S7 · Bridge-dependence on never-rehearsed routes (Table S7). Layer 12, Qwen, seed 0: identity +0.503 (base), +0.545 (none), +0.597 (route), +0.682 (bridge-as-context), +0.602 (atomic); strict donor-following 0.439 / 0.498 / 0.634 / 0.671 / 0.608.

S8 · The design that preceded the within-model split. A first design compared two adapters trained on matched fact sets, one about bridges of measured routes and one about entities in no route. Matching predicate, subject type, conflict class and training tokens did not match perturbation: the control adapter’s subjects were less familiar to the model, harder to learn, and did $3.7\times$ more damage to held-out perplexity at the matched checkpoint, producing an effect in the wrong direction. The within-model split of §3.2 replaced it, so that the comparison of injected and non-injected routes shares one adapter’s general disruption.

S9 · Second-hop knowledge at baseline (Table S11). The route builder does not require the second hop to be answered when asked directly. On the comparison set, the base model answers all six direct second-hop phrasings for 179 of 229 Qwen routes and 231 of 414 Falcon routes, and none for 21 and 117. The primary contrast restricted to the first group is +0.148 [+0.007, +0.289] (Qwen) and +0.175 [+0.047, +0.302] (Falcon); on routes with the second hop known on at least one phrasing, +0.153 [+0.019, +0.287] and +0.183 [+0.067, +0.299]. Requiring in addition that both facts were rehearsed under the atomic condition (§3.3) gives +0.153 [+0.010, +0.296] on 162 Qwen routes; Falcon’s routes are all paired, so its value is unchanged. These analyses were not declared in advance.

S10 · Decision rules of the format ablations (Table S12). For each family, checkpoint and route set: the share of the masked-to-bridge-as-context gap closed by answer-only loss, and bridge-as-context minus answer-only with its paired interval and sign count. Three-seed rows are the declared G17 outcome; six-seed rows are the post hoc extension (G18); the row pooled across families was not declared.

Tables S1–S12 are results/tables/T1_preservation.md through T12_decisions.md.

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Disclosure of LLM use

Design, code, runs, analyses and drafting were carried out with Claude (Anthropic) under the author’s direction, and a second, independent model reviewer examined the design and results at three points. The author directed the work and is responsible for every claim.

Reproducibility

Code, data builders, declared-before-the-run predictions, result files and the table generator are released with this preprint. Model revisions are pinned (`src/model_pin.py`).